MAJOR PROJECT PHASE I · MID TERM EVALUATION

STRATEGIC APPLICATION OF ARTIFICIAL INTELLIGENCE IN CAD

SUBMITTED BY

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GUIDE

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Live demo
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aicad-vercel.vercel.app

PROBLEM FORMULATION

Given a requirement in plain English, produce a verified design of a standard joint

The problem. A knuckle, cotter or coupling joint is sized by a fixed sequence of textbook formulas, then checked against a fixed list of failure modes. A change of load means repeating the whole sequence by hand.

Why a modeller does not solve it. Parametric CAD regenerates whatever dimensions it is given. It does not object if a pin is too slender to survive.

Application and importance. First-pass sizing, checking and drawing of standard joints for a design office or a machine design laboratory, with every number traceable to a page, so the result can be checked rather than trusted.

LITERATURE INTEGRATED	USED FOR
NPTEL IIT Kharagpur, Design of Machine Elements I, Module 4	procedure, proportions, failure modes
Unwin, Elements of Machine Design (1899), p. 179	gib and cotter joint
Leutwiler, Elements of Machine Design (1917), Art. 98, 275	second source; coupling
SP 46, IS 2048, IS 1363/1364, IS 2693	drawing practice, keys, bolts, couplings
OCCT, CadQuery, FreeCAD, CalculiX, gmsh	modelling, drawing, finite element

12 references in the report, 3 of them the design sources every formula traces to.

METHODOLOGY

Five stages; the AI reads the requirement and the engineering stays deterministic

01 · AI
Read
Sentence to joint + load. Nothing invented is kept.

02
Size
Governing formula, rounded to a preferred size.

03
Check
Every failure mode vs allowable; iterate on failure.

04
Model
Exact solids, STEP files, assembly in position.

05
Draw
First angle A3 sheet, dimensions, title block.

Assumptions stated. Static loading only. Allowable tensile 75, shear 60, crushing 150 MPa (working values of the source). Design factor 1.25: sized for 1.25 times the load, checked at the real load. Computed diameters rounded up to a preferred size.

Tools. CadQuery on the OCCT kernel for solids, FreeCAD for the A3 sheet, CalculiX with gmsh for the finite element cross-check, a language model for reading the sentence only.

METHODOLOGY · WHY THE SPLIT, AND THE CODE THAT ENFORCES IT

The AI never picks a dimension: any number not in the request is dropped

```
def _traceable(value, text, factors=(1,)) -> bool:
    for n in _numbers_in(text):
        for f in factors:
            if abs(n * f - value) <= 0.01 * max(abs(value), 1e-9):
                return True
    return False

# used on every field the AI returns:
if not _traceable(v, text, factors):
    spec.notes.append(f"AI gave {key}={v:g}, which is not in the request; dropped")
    continue
```

Reading "50 kN" as 50,000 N is a language problem. Choosing a pin diameter has one correct answer from a formula, so that stays deterministic and traceable to a book page. Nothing anywhere is trained.

INITIAL CALCULATIONS AND RESULTS

Worked example: a knuckle joint for 50 kN, all nine checks at the real load

SIZING, BEFORE ROUNDING	MM	ADOPTED
Rod, direct tension	32.57	35 mm
Pin, double shear	25.75	not governing
Pin, bending	43.72	governs, 45 mm

Bending asks for 1.7 times what shear asks for. A pin sized from shear alone would pass its own check and bend in service. The source takes the same step, and the source's worked example was reproduced value by value (report, Section 4.3).

CHECK	MPA	ALLOW.	USED	%
rod tension	52	75		69%
pin double shear	15.7	60		26%
pin bending	55	75		73%
eye tension	45.7	75		61%
eye shear	45.7	60		76%
eye crushing	25.4	150		17%
fork tension	38.1	75		51%
fork shear	38.1	60		64%
fork crushing	21.2	150		14%

INITIAL CALCULATIONS AND RESULTS · ALL JOINTS

6 fundamental joints, 44 failure checks, every one satisfied

JOINT	PARTS	CHECKS	BUSIEST, % ALLOWABLE
Knuckle joint	5	9	76
Socket and spigot cotter joint	3	10	67
Sleeve and cotter joint	5	8	77
Gib and cotter joint	4	8	79
Hex bolt, nut and washer	3	2	71
Protected type rigid flange coupling	12	7	79

Why these 6. Between them they exercise tension, double shear, crushing, bending and torsion. The flange coupling is sized from torque rather than axial load, to show the method is not limited to one kind of joint. Riveted and welded joints follow as the extension, on the same architecture.

PRESENTATION SKILLS

Five sections, five presenters, each owning what they can defend

Ansh: the problem and the literature. Bhavya: what was built and the AI/engineering boundary. Atishay: the worked example, the output and the code. Dharmendra: the verification and the three findings. Ankur: validation, limits and the close.

On questions. Every claim traces to a file or a source and is said out loud if asked. Nothing is presented as more certain than it was checked to be.

VIVA-VOCE AND FINAL PROJECT REPORT IN SOFT COPY

The report is the evidence file for every answer

50	8	12	44
PAGES, DEPARTMENT FORMAT	CHAPTERS PLUS APPENDIX	REFERENCES	CHECKS, EACH WITH ITS ARITHMETIC

Structured for questions. Chapter 7 records the results, the verification against the sources and the three discrepancies found in them; the soft copy carries the figures and the live-demo QR code.

LIVE DEMO

Try it yourself: scan, or open aicad-vercel.vercel.app

Set the reader to "rules only" (or "AI, else rules"). The hosted copy has no AI key, so "AI only" will report an error.

Sentences that work: knuckle joint for 50 kN · gib and cotter joint for 50 kN · socket and spigot joint for 35 kN · flange coupling to transmit 20 kW at 1440 rpm.

Change the load and every dimension is recomputed, nothing is stored.